Question 1 : Create a Card Visual to show Total Sales.

Objective: Display the total sales amount from the sales dataset using a Card Visual in Power BI.

Step 1: Create a Total Sales Measure

Go to Modeling → New measure and enter the following DAX formula:

```
Total Sales = SUM(SalesData[Total Sales])
```

Note: If your table or sales column has a different name, replace `SalesData[Total Sales]` with the correct names.

Step 2: Create the Card Visual

1. Open Report View in Power BI Desktop.
2. In the Visualizations pane, select the Card visual.
3. Drag the `Total Sales` measure into the Card's Data or Values field.
4. Open the Format visual pane.
5. Customize the card:
    - Title: Total Sales
    - Callout value: Adjust the font size.
    - Display units: Choose Auto, Thousands, or Millions.
    - Decimal places: Set as required.

Expected Output

TOTAL SALES

₹1,25,000

Illustrative value only — your card will display the actual total from your dataset.

Result

The Card Visual displays a single KPI representing the sum of sales across all records. It updates automatically when report filters or slicers are applied.

Question 2 : Create a Donut Chart showing Total Sales by Ship Mode and which ship mode generates the highest sales?

Objective: Create a Donut Chart to compare total sales across different shipping modes and identify which ship mode generates the highest sales.

Step 1: Create a Total Sales Measure

In Power BI Desktop, go to Modeling → New measure and enter:

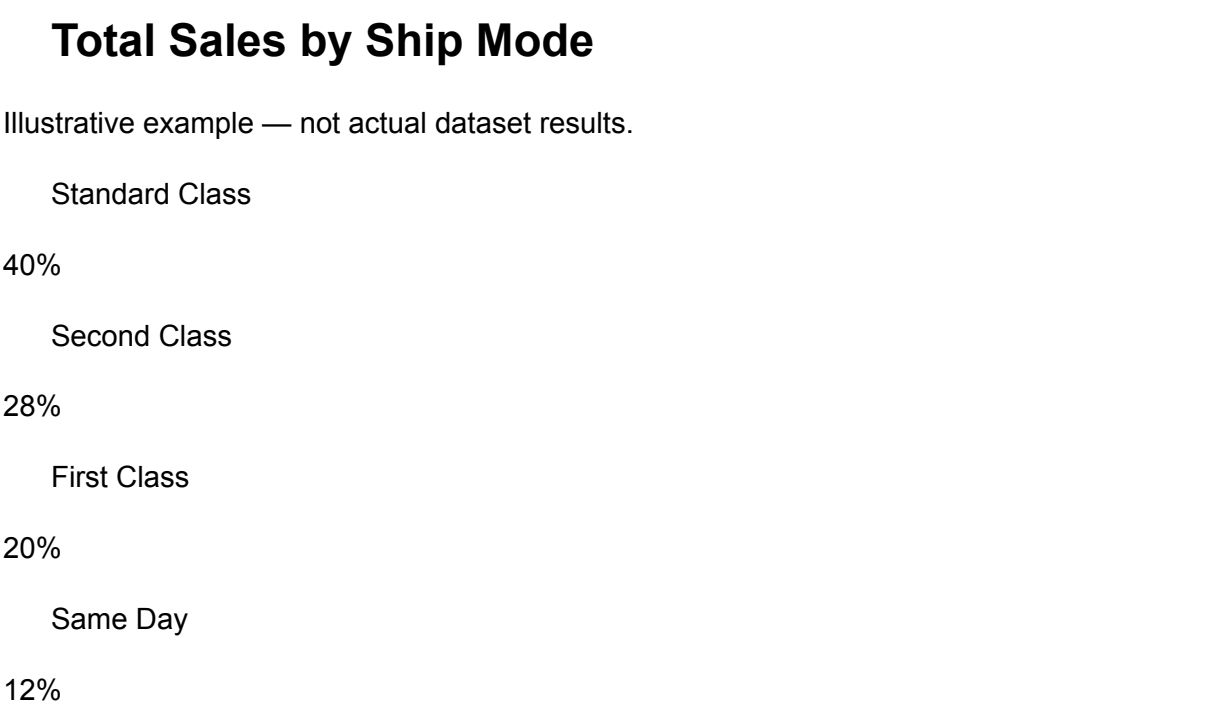
```
Total Sales = SUM(SalesData[Sales])
```

Note: If you already created the `Total Sales` measure in Question 1, you can reuse it. Replace `SalesData` or `Sales` if your dataset uses different names.

Step 2: Create the Donut Chart

1. Open Report View in Power BI.
2. Select the Donut Chart visual from the Visualizations pane.
3. Drag the `Ship Mode` column into Legend.
4. Drag the `Total Sales` measure into Values.
5. Open Format visual and enable:
    - Title: Total Sales by Ship Mode
    - Detail labels: Show category, sales value, and percentage of total, if available.
    - Legend: On.

Step 3: Expected Output



Step 4: Which Ship Mode Generates the Highest Sales?

The ship mode with the largest segment in the Donut Chart generates the highest total sales.

In the illustrative example above, Standard Class generates the highest sales at 40%.

Conclusion: Check your actual Donut Chart to identify the ship mode with the highest sales. The exact answer depends on your dataset.

Question 3 : Create a Line Chart showing the Sales Trend over time (by Month and Year). Is there any noticeable seasonal pattern in sales?

Objective: Create a Line Chart to analyze monthly sales trends and identify any seasonal patterns in sales.

Step 1: Create a Total Sales Measure

Go to Modeling → New measure and enter:

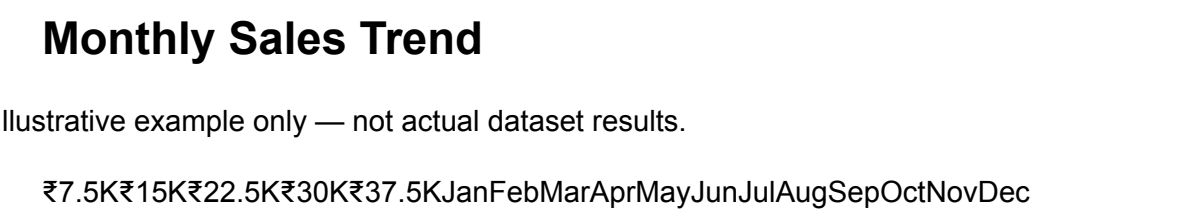
```
Total Sales = SUM(SalesData[Sales])
```

If you already created this measure in Question 1 or 2, reuse it.

Step 2: Create the Line Chart

1. Open Report View in Power BI Desktop.
2. Select the Line Chart visual from the Visualizations pane.
3. Drag the `Order Date` column to the X-axis.
4. Select the date field's hierarchy and use Year and Month, or use a proper calendar table.
5. Drag the `Total Sales` measure to the Y-axis.
6. In the Format visual pane, set the title to Monthly Sales Trend.
7. Ensure months are sorted chronologically, from January to December, within each year.

Step 3: Expected Output



Illustrative example only — not actual dataset results.

Example monthly sales pattern

Step 4: Is There Any Noticeable Seasonal Pattern in Sales?

Seasonality means sales repeatedly increase or decrease during particular months or periods of the year.

Look for these patterns in your actual chart:

- Peak sales: Months that consistently show higher sales.
- Low sales: Months that repeatedly show lower sales.
- Seasonal increases: Sales may rise during festivals, holidays, or year-end shopping periods.
- Year-over-year patterns: Compare the same months across different years to see whether the pattern repeats.

Conclusion

The Line Chart helps identify sales growth, declines, monthly fluctuations, and potential seasonal trends. In the illustrative chart, sales increase toward November and December, suggesting a possible year-end seasonal peak.

Note: You need to examine your actual dataset to confirm whether seasonality exists. A single year's increase does not establish a recurring seasonal pattern.

Do you like this personality?

Question 4 : Create a Bar Chart to compare Sales by Category.

Objective: Create a Bar Chart to compare total sales across different product categories and identify which category generates the highest sales.

Step 1: Create a Total Sales Measure

In Power BI Desktop, go to Modeling → New measure and enter:

```
Total Sales = SUM(SalesData[Sales])
```

If you have already created the `Total Sales` measure, reuse it.

Step 2: Create the Bar Chart

1. Open Report View in Power BI.
2. Select the Clustered Bar Chart visual from the Visualizations pane.
3. Drag the `Category` column into the Y-axis.
4. Drag the `Total Sales` measure into the X-axis.
5. Open Format visual and:
    - Set the title to Sales by Category.
    - Turn on Data labels.

- Sort by `Total Sales` in descending order.

Step 3: Expected Output

Sales by Category

Illustrative values only — your actual chart will depend on your dataset.

Technology

₹45,000

Furniture

₹32,000

Office Supplies

₹25,000

Step 4: Interpretation

- Highest sales: The category with the longest bar generates the most sales.
- Lowest sales: The category with the shortest bar generates the least sales.
- Comparison: The chart helps identify strong and weak product categories for business planning.

Conclusion

The Bar Chart provides a clear comparison of total sales by category. In the illustrative example, Technology has the highest sales, followed by Furniture and Office Supplies.

Note: Check your actual Power BI chart to determine which category generates the highest sales in your dataset.

Question 5 : Create a Matrix Visual to display Sales by Month and Category and identify the month with the highest sales.

Objective: Display total sales by Month and Category, then identify the month with the highest sales.

Step 1: Create a Total Sales measure

In Power BI, go to Modeling → New measure and enter:

```
Total Sales = SUM('SalesData'[Sales])
```

Replace `'SalesData'` with your actual table name if it is different.

Step 2: Create the Matrix visual

Category	April	February	January	June	March	May	Total
Office Furniture	**$106,529**	**$86,712**	**$71,717**	**$89,376**	**$90,573**	**$85,796**	**$530,703**
Cabinets & Cupboards	**$43,687**	**$33,087**	**$40,695**	**$43,850**	**$45,166**	**$55,368**	**$261,853**
File Cabinets	$14,510	$17,896	$16,006	$18,842	$22,759	$25,199	**$115,212**
Bookcases	$29,177	$15,191	$24,689	$25,008	$22,407	$30,169	**$146,641**
Chairs	**$28,737**	**$21,806**	**$15,475**	**$20,721**	**$24,786**	**$15,724**	**$127,249**
Office Chair	$28,737	$21,806	$15,475	$20,721	$24,786	$15,724	**$127,249**
Workstations	**$34,105**	**$31,819**	**$15,547**	**$24,805**	**$20,621**	**$14,704**	**$141,601**
Office Desk	$34,105	$31,819	$15,547	$24,805	$20,621	$14,704	**$141,601**
Office Supplies	**$143,565**	**$137,716**	**$152,005**	**$138,920**	**$123,151**	**$175,537**	**$870,894**
Paper	**$76,968**	**$80,012**	**$76,866**	**$88,439**	**$82,172**	**$88,482**	**$492,939**
Photo Paper	$17,458	$8,258	$21,635	$19,440	$23,127	$18,711	**$108,629**
Memo & Scratch Pads	$22,088	$29,553	$17,837	$18,163	$22,943	$22,583	**$133,167**
Copy & Multipurpose Paper	$14,895	$12,219	$20,480	$27,354	$17,757	$18,376	**$111,081**
Cards	$22,527	$29,982	$16,914	$23,482	$18,345	$28,812	**$140,062**
Stationery	**$66,597**	**$57,704**	**$75,139**	**$50,481**	**$40,979**	**$87,055**	**$377,955**
Pencil	$24,523	$18,221	$18,415	$16,545	$13,800	$21,232	**$112,736**
Pen	$25,588	$12,686	$28,938	$12,236	$14,772	$28,043	**$122,263**
Highlighter	$16,486	$26,797	$27,786	$21,700	$12,407	$37,780	**$142,956**
Total	**$250,094**	**$224,428**	**$223,722**	**$228,296**	**$213,724**	**$261,333**	**$1,401,597**

1. Select Matrix

In the Visualizations pane, click the Matrix visual icon.

Sub-Category

- ☐ Cabinets & Cupboards
- ☐ Chairs
- ☐ Paper
- ■ Stationery
- ☐ Workstations

Category	April	February	January	June	March	May	Total
⊟ **Office Supplies**	**$66,597**	**$57,704**	**$75,139**	**$50,481**	**$40,979**	**$87,055**	**$377,955**
⊟ **Stationery**	**$66,597**	**$57,704**	**$75,139**	**$50,481**	**$40,979**	**$87,055**	**$377,955**
Pencil	$24,523	$18,221	$18,415	$16,545	$13,800	$21,232	**$112,736**
Pen	$25,588	$12,686	$28,938	$12,236	$14,772	$28,043	**$122,263**
Highlighter	$16,486	$26,797	$27,786	$21,700	$12,407	$37,780	**$142,956**
Total	**$66,597**	**$57,704**	**$75,139**	**$50,481**	**$40,979**	**$87,055**	**$377,955**

2. Add the fields

- Rows: Month (from Order Date)
- Columns: Category
- Values: Total Sales

If you only have an Order Date field, use its Month level in the date hierarchy or create a Month column.

Step 3: Format the Matrix

- Turn on Row subtotals and Column grand total to compare monthly and category totals.
- Apply conditional formatting with a color scale to highlight higher sales.
- Sort Month chronologically, not alphabetically. If the data spans multiple years, include Year as well.



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Step 4: Identify the month with the highest sales

Create this optional measure to return the highest monthly sales value:

```
- Highest Monthly Sales =
- MAXX(
-     SUMMARIZE(
-         'SalesData',
-         'SalesData'[Month],
-         "MonthlySales", [Total Sales]
-     ),
-     [MonthlySales]
)
```

This measure assumes you have a Month column and evaluates each month across the dataset. If multiple years are present, use a Year-Month field to distinguish the same month in different years.

Conclusion / interpretation for your assignment:

“The Matrix visual displays total sales for each month across different product categories. By comparing the monthly totals, we can identify the month with the highest sales and determine which categories contribute most to revenue. The exact month cannot be identified without the actual dataset.”

For the official Power BI instructions, see [Microsoft Learn: Create a Matrix Visual](#).

Question 6 : Create a Decomposition Tree for Sales Breakdown.

Objective: Analyze Total Sales by Region, Category, Sub-Category, and Product to identify the factors contributing most to sales. A Decomposition Tree allows you to drill down into different dimensions and explore the reasons behind a sales result.



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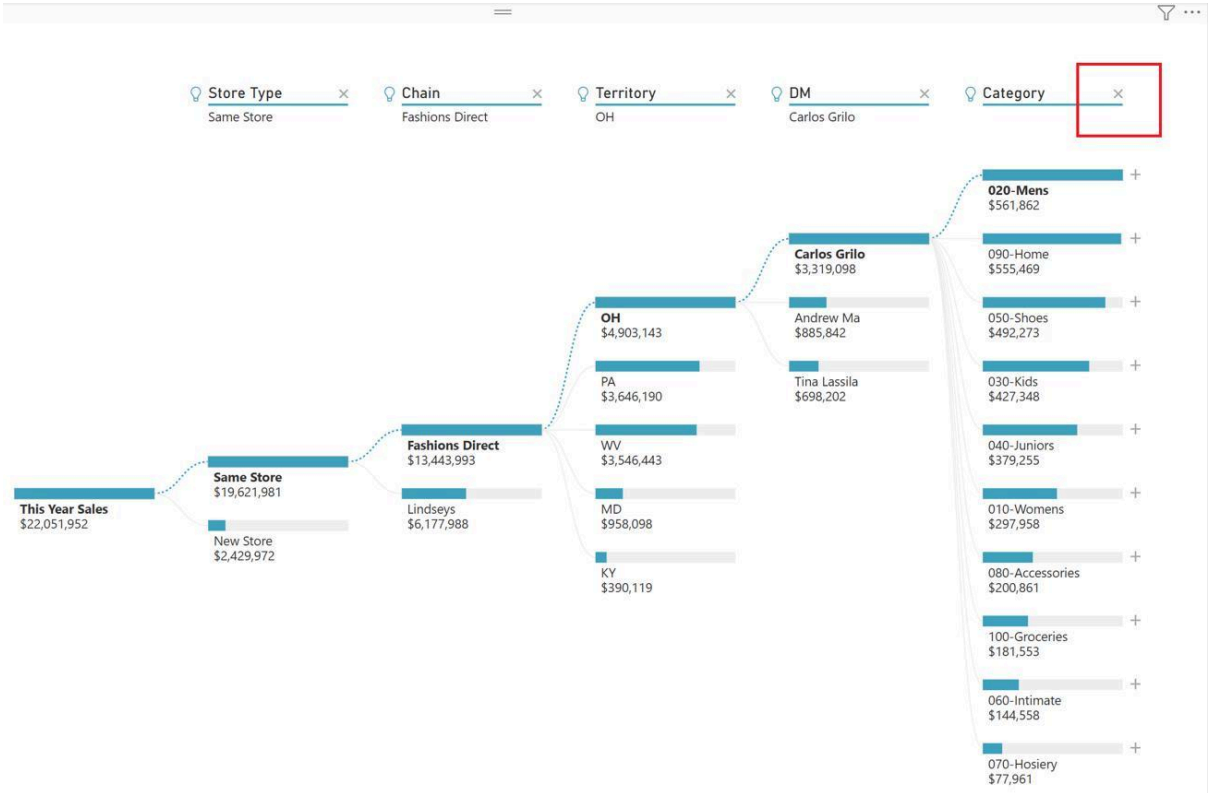
Step 1: Create the Total Sales measure

Go to Modeling → New measure and enter:

```
Total Sales = SUM('SalesData'[Sales])
```

Replace `'SalesData'` with your actual table name.

Step 2: Insert the Decomposition Tree



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1. Open your report in Power BI Desktop.
2. Select the Decomposition Tree icon in the Visualizations pane.
3. Resize the visual to fit your report page.

Step 3: Add the fields

In the Build visual pane, configure the fields as follows:

Analyze

Total Sales

Explain By

Region
Category
Sub-Category
Product Name

Step 4: Analyze the sales breakdown

1. Select the + sign next to Total Sales.
2. Choose Region to compare sales across regions.
3. Select the + sign on a region and choose Category.
4. Continue drilling into Sub-Category and Product Name.

5. Alternatively, choose High value from the AI split options to find the dimension and value associated with the highest sales.



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Step 5: Conclusion for your assignment

The Decomposition Tree visual helps identify which regions, product categories, sub-categories, and individual products contribute most to total sales. It supports interactive exploration and helps businesses identify their strongest sales drivers.

Expected finding: The highest-sales region, category, and product can be identified by expanding the tree and selecting High value. The exact results depend on your dataset.

For official instructions, refer to Microsoft Learn: Create and View Decomposition Tree Visuals.

Question 7 : Create a Scatter Chart to compare Discount vs. Profit by Sub-Category. Are there any outliers where high discounts resulted in negative profit?

Objective: Identify the relationship between discounts and profit across product sub-categories and find outliers where high discounts are associated with negative profit.

A scatter chart is suitable because it displays the relationship between two numerical variables and helps identify patterns, clusters, and outliers.



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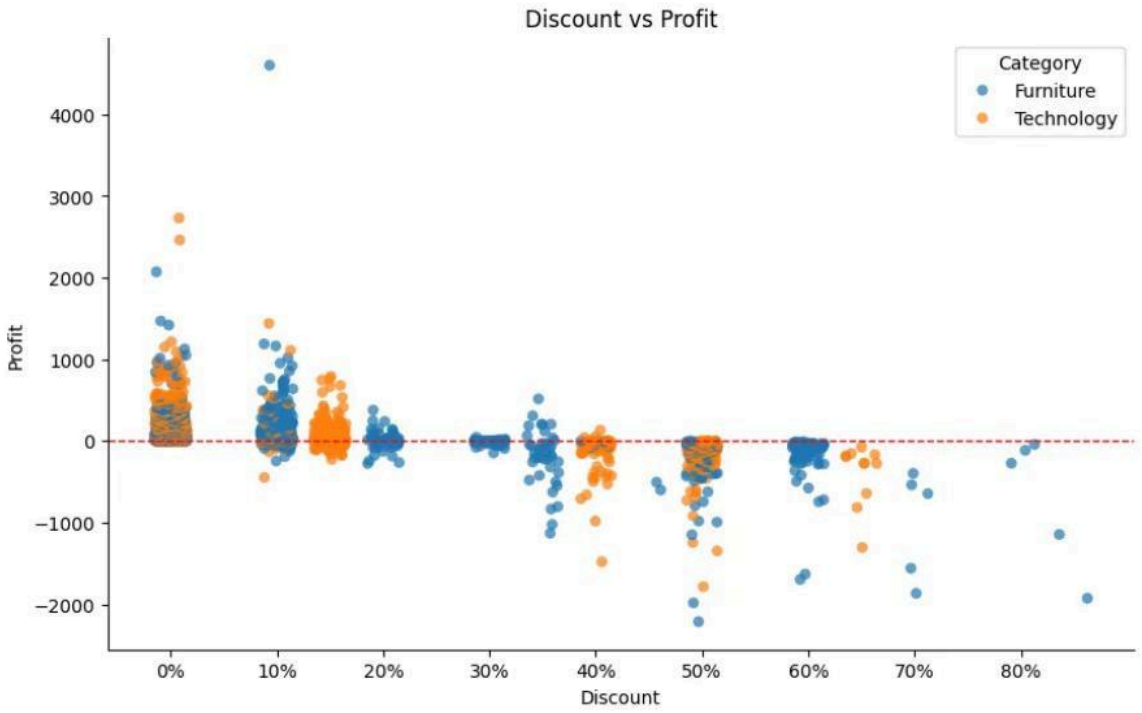
Step 1: Create the required measures

Go to Modeling → New measure and create:

```
Total Profit = SUM('SalesData'[Profit])
Average Discount = AVERAGE('SalesData'[Discount])
```

Replace `'SalesData'` with your actual table name.

Step 2: Insert and configure the Scatter Chart



dataquestio.medium.com

In Power BI Desktop:

1. Select the Scatter chart visual.
2. Assign the following fields in the Build visual pane.

Visual field	Column or measure
X-axis	Average Discount
Y-axis	Total Profit
Values / Details	Sub-Category, if available in your version
Legend	Category

Size	Sales (optional)

Each point should represent a sub-category. Check the aggregation and field settings to ensure you see one point per sub-category rather than one aggregated point for the entire dataset.

Step 3: Add a zero-profit reference line

In the Format visual or Analytics options, add a horizontal constant line at Profit = 0 if supported. Points below this line represent loss-making sub-categories.

Step 4: Interpret the results

For a commonly used Sample Superstore dataset, published analysis reports these loss-making sub-categories:



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Sub-Category	Total Profit	Interpretation
Tables	-$17,725	Significant losses
Bookcases	-$3,473	Loss-making
Supplies	-$1,189	Loss-making

A separate analysis of the same dataset found that profit margins became negative in discount bands above 20%, with the greatest losses in the highest-discount band.



GitHub

Conclusion for your assignment

Yes, high discounts can be associated with negative profit. The scatter chart can reveal outliers below zero profit, particularly among heavily discounted products. In the commonly used Sample Superstore dataset, Tables and Bookcases are notable loss-making sub-categories. However, confirm the exact points in your own chart because aggregated sub-category results can differ from individual transaction results.

Official reference: [Microsoft Learn — Scatter Charts in Power BI]

Question 8 : Use a Funnel Chart to show the number of orders by Customer Segment.

Objective: Display the number of orders placed by each customer segment, such as Consumer, Corporate, and Home Office, using a Funnel Chart in Power BI.



Microsoft Learn

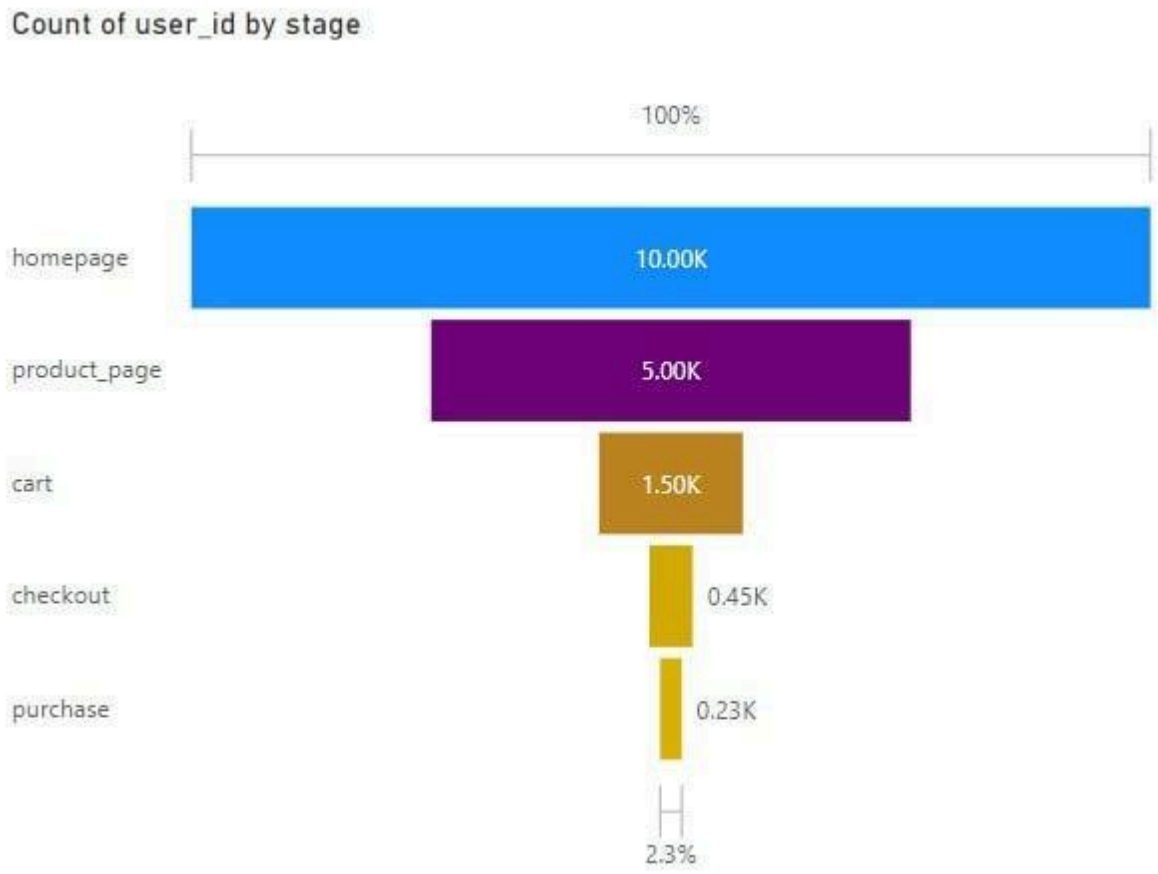
Step 1: Create a measure for the number of orders

In Power BI Desktop, go to Modeling → New measure and enter:

```
Total Orders = DISTINCTCOUNT('SalesData'[Order ID])
```

Replace `'SalesData'` and `[Order ID]` with the names used in your dataset. `DISTINCTCOUNT` counts each unique order once, even if an order contains multiple product rows.

Step 2: Insert the Funnel Chart



excelmojo.com

1. Open the Visualizations pane.
2. Select the Funnel Chart icon.
3. Add the following fields.

Funnel field	**Column or measure**
Category	Customer Segment
Values	Total Orders

Power BI will display the order count for each customer segment. You can also sort the visual in descending order to make the largest segment easier to identify.



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Step 3: Format the visual

- Title: Number of Orders by Customer Segment
- Data labels: On
- Category labels: On
- Sort: Total Orders, descending

Step 4: Interpret the result

The Funnel Chart helps compare the number of orders across customer segments and identify which segment contributes the most orders.

Conclusion for your assignment:

“The Funnel Chart visualizes the distribution of orders across Consumer, Corporate, and Home Office segments. The segment with the largest funnel section has the highest number of orders, while the smallest section has the lowest. The exact leading segment depends on the dataset.”

Official reference: [Microsoft Learn — Create and Use Funnel Charts in Power BI]

Question 9 : Use a Waterfall Chart to analyze how Profit changes by Category within the countries

Objective: Create a Waterfall Chart in Power BI to analyze how each product category contributes to profit within different countries, highlighting positive and negative profit contributions.

A Waterfall Chart displays a running total as values increase or decrease, helping you understand how different categories affect overall profit.



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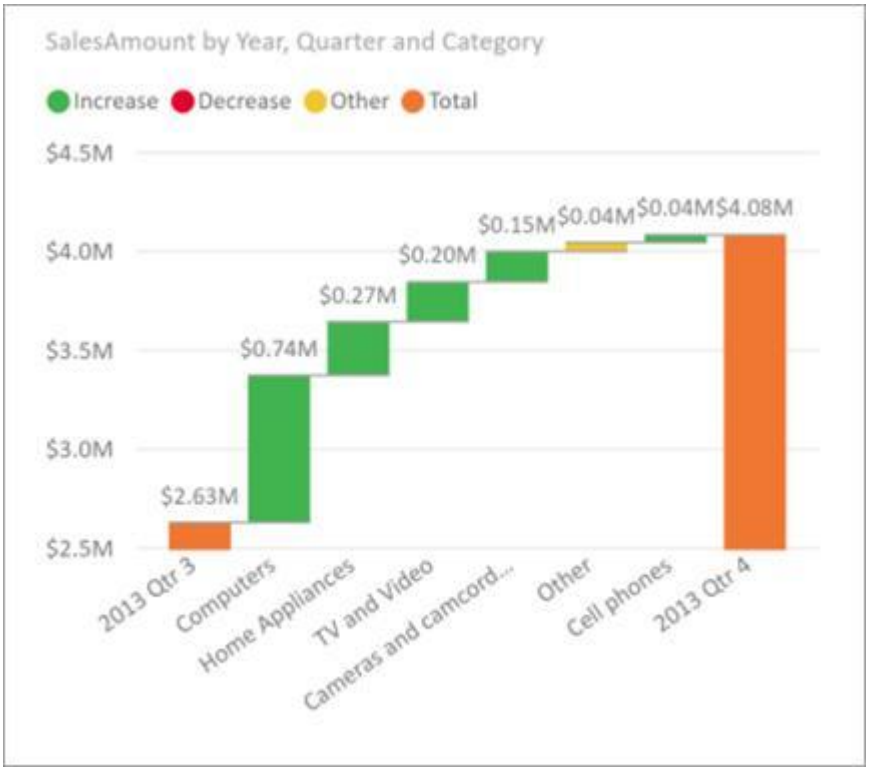
Step 1: Create a Total Profit measure

Go to Modeling → New measure and enter:

```
Total Profit = SUM('SalesData'[Profit])
```

Replace `'SalesData'` with your actual table name.

Step 2: Insert and configure the Waterfall Chart



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In Power BI Desktop:

1. Select the Waterfall Chart visual from the Visualizations pane.
2. Assign the following fields:

Visual field	Column or measure
Category	Category
Y-axis / Values	Total Profit
Breakdown	Country

Tooltips	Country, Category, Total Profit

The chart shows profit by product category, with country-level contributions within each category.



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Step 3: Analyze the results

- Positive profit: The category contributes positively to profit.
- Negative profit: The category reduces profit or contributes to a loss.
- Country breakdown: Helps identify which countries contribute most to increases or decreases.
- Sort: Use the chart's More options menu to sort by profit or category.

Step 4: Conclusion for your assignment

“The Waterfall Chart illustrates how different product categories affect total profit across countries. Positive contributions indicate profitable categories, while negative contributions highlight areas of loss. This analysis helps businesses identify their strongest markets and investigate categories that reduce profitability.”

Note: The exact country and category with the highest or lowest profit depend on your dataset.

Official reference: [Microsoft Learn — Waterfall Charts in Power BI]

Question 10 : Create a Dual-Axis Line Chart to show Total Sales and Profit Over Time (by Month).

Objective: Compare monthly Total Sales and Total Profit in Power BI using two Y-axes, making it easier to observe trends when sales and profit have different value ranges.



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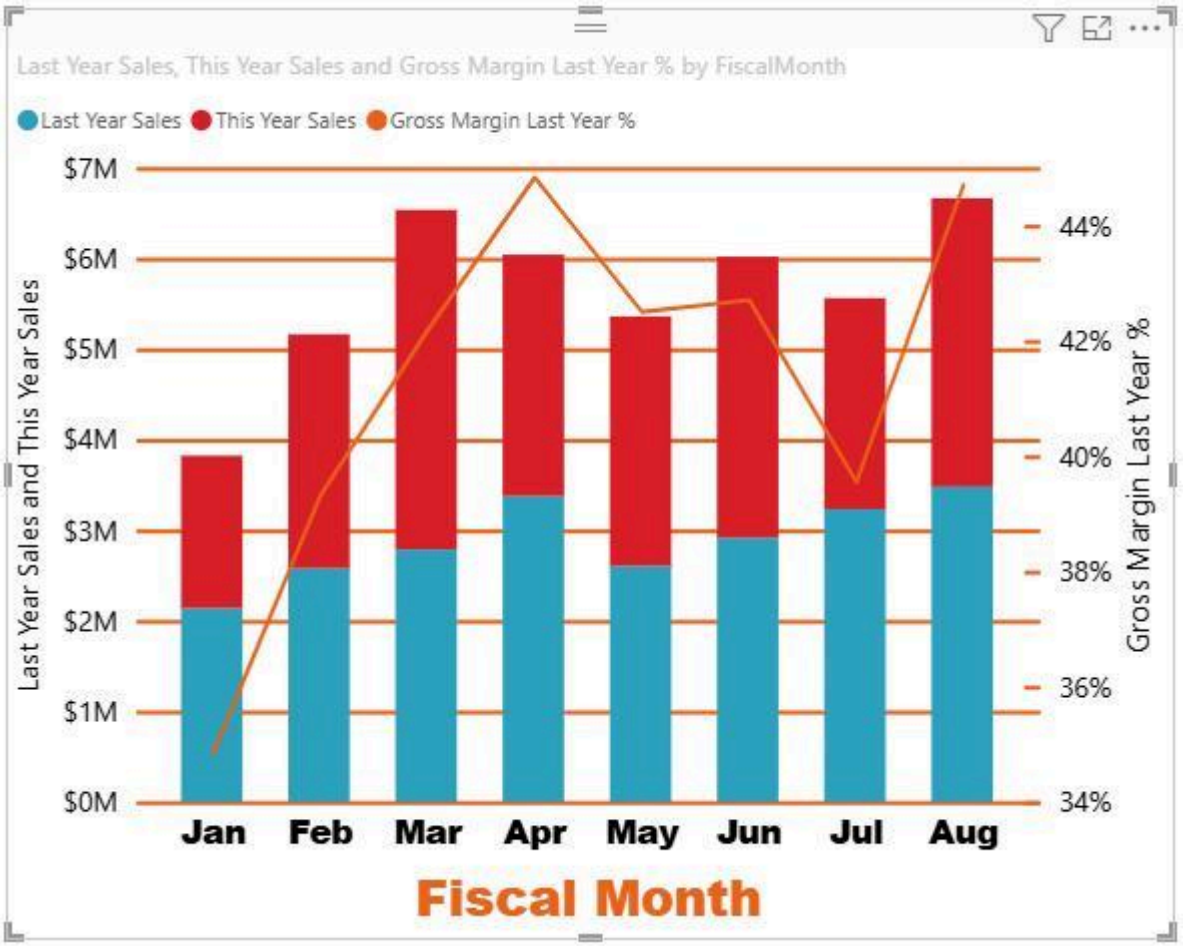
Step 1: Create the measures

Go to Modeling → New measure and create these measures:

```
Total Sales = SUM('SalesData'[Sales])
Total Profit = SUM('SalesData'[Profit])
```

Replace `'SalesData'` with your actual table name.

Step 2: Create the chart



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Power BI's standard combo chart supports a secondary axis, but it combines columns and a line rather than displaying two separate lines. To create a true dual-axis chart with two lines, use a suitable custom visual if available. Alternatively, use the built-in combo chart as follows.

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1. Select Line and clustered column chart in the Visualizations pane.
2. Drag the following fields into the visual:

Visual field	Value
X-axis / Shared axis	Order Date → Month
Column Y-axis	Total Sales
Line Y-axis	Total Profit

3. If your dataset spans multiple years, use a Year-Month field to keep months in chronological order.

Step 3: Enable the secondary Y-axis

1. Select the chart and open Format visual.
2. Expand Y-axis and Secondary y-axis.
3. Turn on the secondary axis and its title.
4. Label the left axis Total Sales and the right axis Total Profit.
5. Give the chart the title Monthly Sales and Profit Trend.



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Step 4: Interpret the chart

- Sales increase and profit increases: Sales growth is accompanied by higher profit.
- Sales increase but profit decreases: Costs, discounts, or changes in product mix may be reducing profitability.
- Both decrease: The business may be experiencing a weaker sales period.
- Seasonal peaks: Repeated rises or falls in similar months may indicate seasonality.

Conclusion for your assignment

“The dual-axis chart compares monthly Total Sales and Total Profit over time. The secondary axis makes both measures easier to interpret when their numerical ranges differ. The chart helps identify growth trends, periods of declining profitability, and potential seasonal patterns. The actual trends must be confirmed using the dataset.”

Official reference: [Microsoft Learn — Create and Use Combo Charts in Power BI]